

Single Phase Induction Motor

A single-phase induction motor is a small-size motor with a fractional-kilowatt rating. They work on the principle of electromagnetic induction to create a rotating magnetic field. It is used in domestic appliances like fans, hair dryers, washing machines, vacuum cleaners, mixers, refrigerators, food processors and kitchen equipment employ these motors.

Construction of Single-Phase Induction Motor

To construct Single Phase Induction Motor, it comprises of two major components which is the rotor and the stator.

Stator: As the name implies, a stator is a stationary component of an induction motor. The stator of a single phase induction motor receives a single phase alternating current source.

Rotor: The rotor is a rotating component of an induction motor. The rotor transmits mechanical load via the shaft. The squirrel cage rotor is used in the single-phase induction motor.

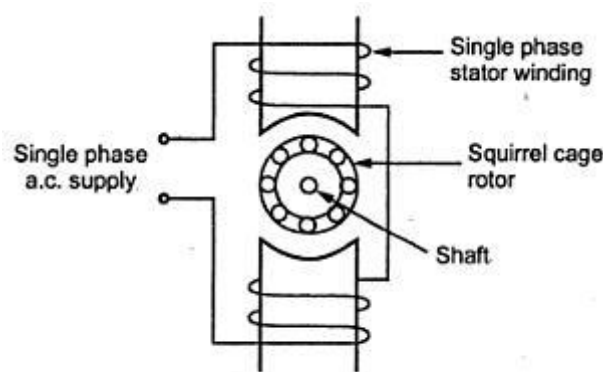


Fig. .2

Operation and Working of Single-Phase Induction Motor

A single-phase induction motor is similar in construction to that of a polyphase induction motor with the difference that its stator has only one winding. When a single phase AC supply is applied to the stator winding of single phase induction motor, the alternating current starts flowing through the stator or main winding. The flux is then generated by the AC current. The flux also links with the rotor conductors and hence cut the conductors of the rotors.

According to the Faraday's law of electromagnetic induction, emf gets induced in the rotor. The current begins to flow in the rotor after the rotor circuit is closed. The flux created by this rotor current is known as the rotor flux. Since this flux is produced due to the induction principle so, the motor working on this principle got its name as an induction motor. There are currently two fluxes: the main flux and the rotor flux. The desired torque, which the motor needs to rotate, is produced by these two fluxes.

Its motors consist of a single-phase winding mounted on the stator and a cage winding on the rotor. When a single-phase supply connected to stator winding pulsating magnetic field is produced. Under these conditions, the rotor does not rotate. It requires some special starting methods.

Why Single Phase Induction Motor is not Self Starting?

A single-phase induction motor lacks a naturally rotating magnetic field, which makes it non-self-starting. Unlike three-phase motors that generate a rotating magnetic field with three alternating currents, single-phase motors rely on a single alternating current that produces a

pulsating magnetic field. This field is unable to initiate continuous rotation in the motor's rotor due to its non-uniform nature and inability to provide sufficient starting torque.

At the starting condition of the motor, **both ϕ_f (forwarding component of alternating flux(ϕ_m)) and ϕ_b (backward component of alternating flux(ϕ_m))** are equal in magnitude but opposite in direction. They cancel each other out, which results zero net torque on the rotor. This zero torque at the starting condition is why single-phase induction motors are not self-starting.

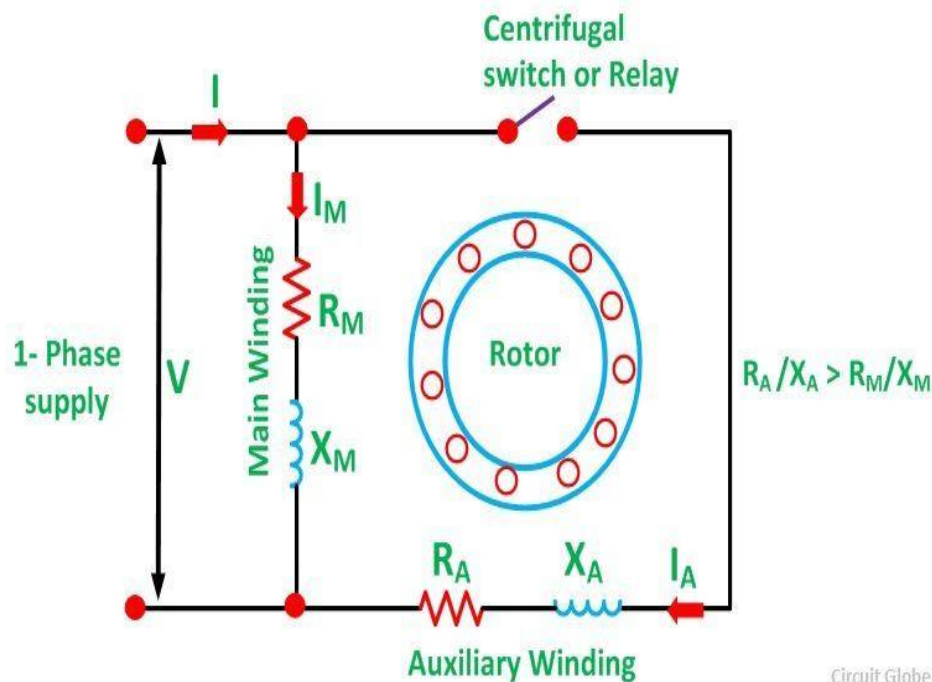
To overcome this limitation, various techniques are used, such as adding an auxiliary winding and a capacitor to simulate a rotating magnetic field during start up or using centrifugal switches to disconnect the starting winding once the motor reaches a certain speed. These methods help single-phase induction motors achieve the necessary torque for self-starting and sustained operation.

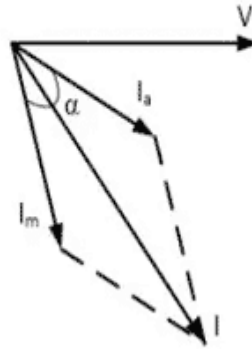
Starting Methods and Types of Single-Phase Induction Motor

1. Split-phase motor
2. Capacitor-start motor
3. Capacitor-start capacitor-run motor (or two-value capacitor motor)

1. Split-Phase Induction Motor

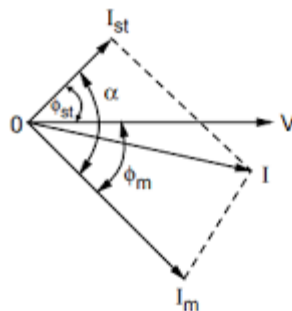
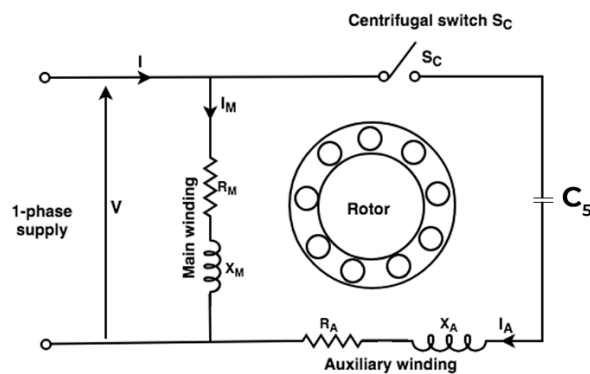
It is also called a resistance start motor. It has a single-cage rotor and its stator has two windings- a main winding and a starting (auxiliary) winding which is displaced 90° in space. The main winding has very low resistance and high inductive reactance.





2. Capacitor-Start Motor

It has a cage rotor and its stator has two winding (main winding and auxiliary winding) which are displaced 90° in space. The capacitor C_s is connected in series with starting windings. The centrifugal switch S_c is also connected.



(b) Phasor diagram

3. Capacitor-Start Capacitor-Run Motor (Two Value Capacitor Motor):-

It has a cage rotor and its stator has two windings (main winding and auxiliary winding) displaced by 90° in space. The motor uses two capacitors C_s (starting capacitor) and C_R (run capacitor). The two capacitors are connected in parallel at the start. To obtain a high starting

torque, a large current is required therefore the capacitive reactance X in the starting torque should be low.

